

# Companion to Bird & de Moor, Algebra of Programming

## §10. Greedy Algorithms

### §10.1. Theory

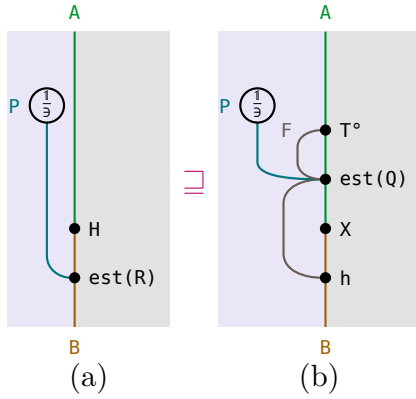
(10.1a)

$h, T, R, H, M$  as in  $\triangle H$ ; additionally  $Q$  a **connected** preorder on the sets  $\frac{T^\circ}{\exists}$  returns, so that  $\frac{T^\circ}{\exists} \text{est}(Q)$  is entire.

**Map(h)  $\wedge$  MonoAlg(h,R)  $\wedge$  R R  $\subseteq$  R  $\wedge$  ThinCondition(T,h,R,Q)  $\Rightarrow$  ( $\mu X : \frac{T^\circ}{\exists} \text{est}(Q) F(X)$  Theorem 10.1  
 $h) \subseteq \frac{H}{\exists} \text{est}(R)$**

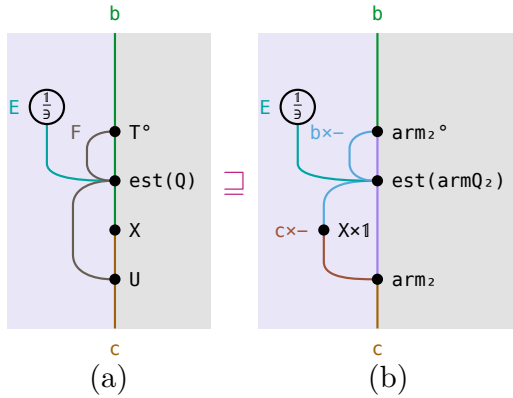
the same optimum reached by keeping ONE decomposition at each step, so that no set is ever carried  
 and the recursion runs on values alone

$H \triangle (T)^\circ (h) \triangle H, Q$



|     |                                                            |  |
|-----|------------------------------------------------------------|--|
| (a) | $\frac{H}{\exists} \text{est}(R)$                          |  |
| (b) | $\sqsubseteq \frac{T^\circ}{\exists} \text{est}(Q) F(X) h$ |  |

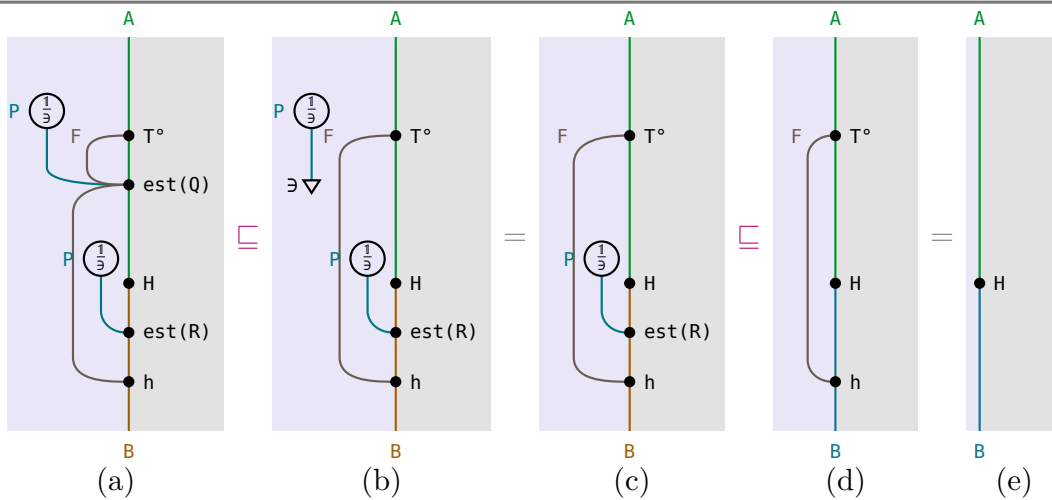
the body at  $T$  a coproduct of two arms



|     |                                                                                                     |                                               |
|-----|-----------------------------------------------------------------------------------------------------|-----------------------------------------------|
| (a) | $\frac{T^\circ}{\exists} \text{est}(Q) F(X) U$                                                      | Proposition 10.1, $V_2 V_1^\circ = \emptyset$ |
| (b) | $\sqsubseteq \frac{\text{arm}_2^\circ}{\exists} \text{est}(\text{arm}_2) (X \times 1) \text{arm}_2$ |                                               |

**$T^\circ F(H) h = H \Rightarrow \frac{T^\circ}{\exists} \text{est}(Q) F(\frac{H}{\exists} \text{est}(R)) h \subseteq H$**  (10.1b)

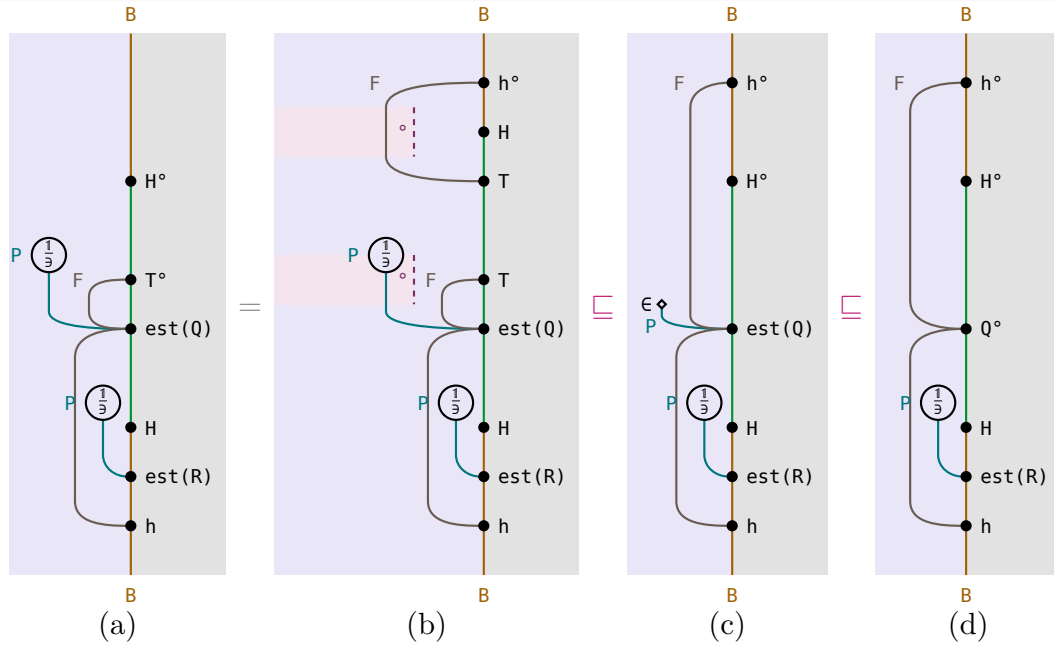
taking the input apart in one  $Q$ -extreme way, solving the parts by  $M$  and assembling by  $h$  returns only what  $H$   
 returns



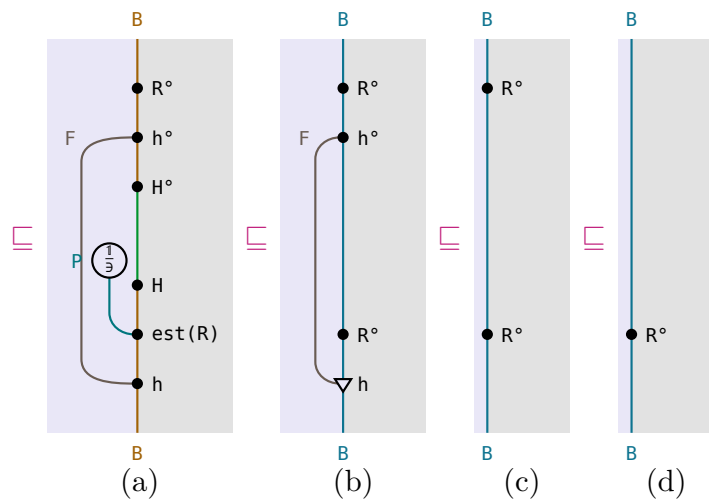
|     |                                                                                    |                                               |
|-----|------------------------------------------------------------------------------------|-----------------------------------------------|
| (a) | $\frac{T^\circ}{\exists} \text{est}(Q) F(\frac{H}{\exists} \text{est}(R)) h$       | $\text{est}(Q) \subseteq \exists$             |
| (b) | $\sqsubseteq \frac{T^\circ}{\exists} \exists F(\frac{H}{\exists} \text{est}(R)) h$ | $\frac{R}{\exists} \exists = R$               |
| (c) | $= T^\circ F(\frac{H}{\exists} \text{est}(R)) h$                                   | $\frac{R}{\exists} \text{est}(R) \subseteq S$ |
| (d) | $\sqsubseteq T^\circ F(H) h$                                                       | $T^\circ F(H) h = H$                          |
| (e) | $= H$                                                                              |                                               |

$$\text{Map}(h) \wedge \text{MonoAlg}(h, R^\circ) \wedge R^\circ R^\circ \subseteq R^\circ \wedge T^\circ F(H) h = H \wedge Q F(H) h \subseteq F(H) h R \implies H^\circ \frac{T^\circ}{3} \text{est}(Q) F(\frac{H}{3} \text{est}(R)) h \subseteq R^\circ \quad (10.1c)$$

$H^\circ$  followed by the greedy body at  $M$  is  $\subseteq R^\circ$ : an answer of the body is never worse than an answer of  $H$  to the same input



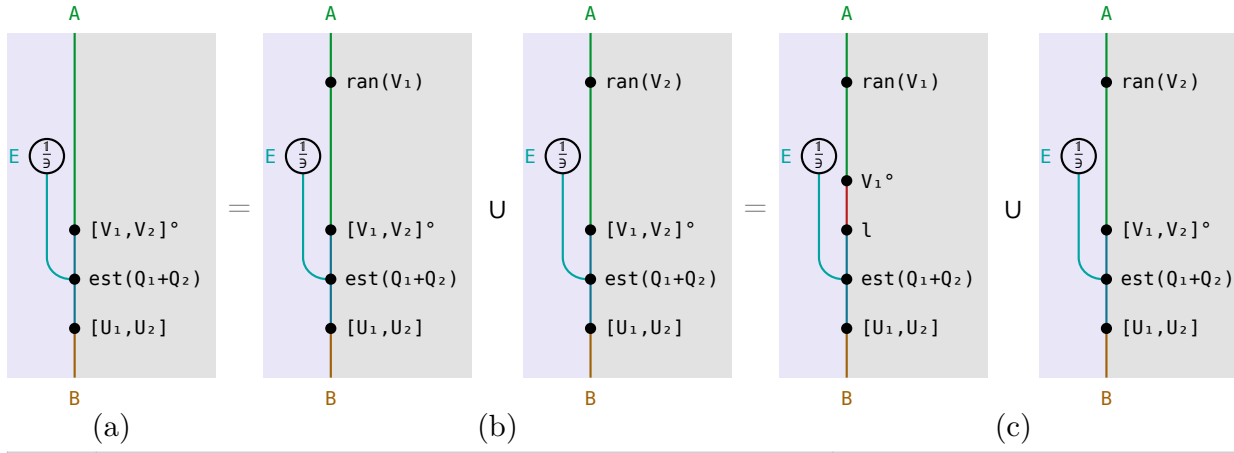
|     |                                                                                         |                                                                                                                                     |
|-----|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $H^\circ \frac{T^\circ}{3} \text{est}(Q) F(\frac{H}{3} \text{est}(R)) h$                | $T^\circ F(H) h = H$<br>$\implies H^\circ = h^\circ F(H^\circ) T$                                                                   |
| (b) | $= h^\circ F(H^\circ) T \frac{T^\circ}{3} \text{est}(Q) F(\frac{H}{3} \text{est}(R)) h$ | $T \frac{T^\circ}{3} \subseteq \epsilon$                                                                                            |
| (c) | $\subseteq h^\circ F(H^\circ) \epsilon \text{est}(Q) F(\frac{H}{3} \text{est}(R)) h$    | $\epsilon \text{est}(R) \subseteq R^\circ$                                                                                          |
| (d) | $\subseteq h^\circ F(H^\circ) Q^\circ F(\frac{H}{3} \text{est}(R)) h$                   | $Q F(H) h \subseteq F(H) h R$<br>$\implies h^\circ F(H^\circ) Q^\circ \subseteq R^\circ h^\circ F(H^\circ)$<br>$t(f g) = t(f) t(g)$ |



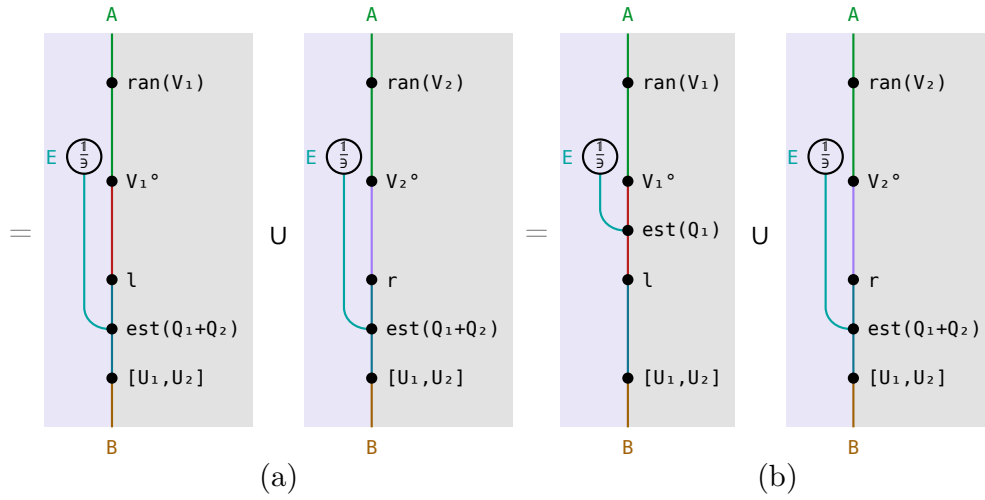
|     |                                                                    |                                                       |
|-----|--------------------------------------------------------------------|-------------------------------------------------------|
| (a) | $\subseteq R^\circ h^\circ F(H^\circ \frac{H}{3} \text{est}(R)) h$ | $S^\circ \frac{S}{3} \text{est}(R) \subseteq R^\circ$ |
| (b) | $\subseteq R^\circ h^\circ F(R^\circ) h$                           | $\text{MonoAlg}(h, R^\circ)$                          |
| (c) | $\subseteq R^\circ R^\circ$                                        | $R^\circ R^\circ \subseteq R^\circ$                   |
| (d) | $\subseteq R^\circ$                                                |                                                       |

$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2] = (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1) \ U_1) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_2) \ U_2) \quad \text{Proposition 10.1}$$

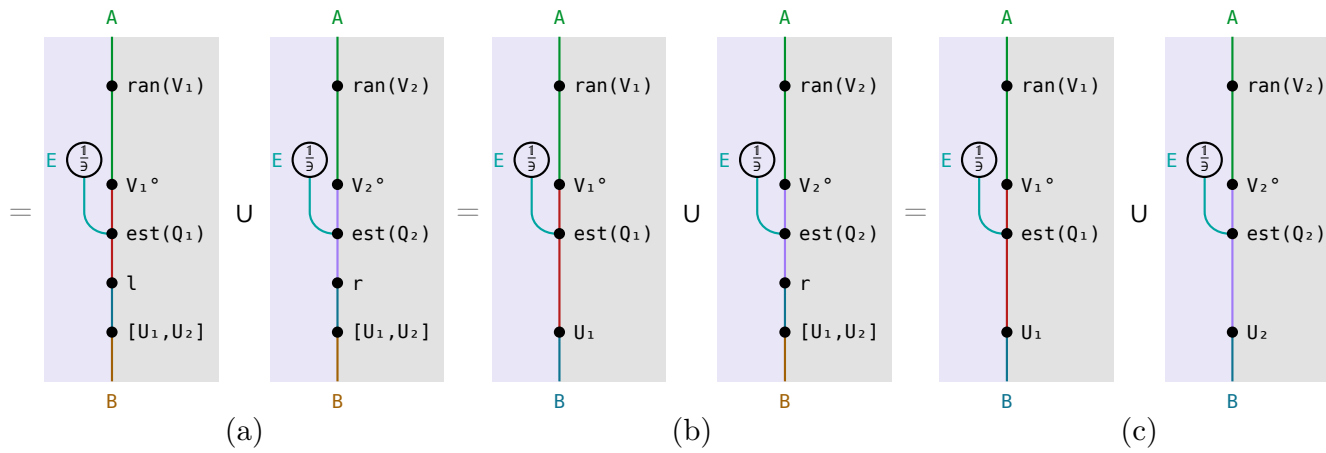
when  $V_1$  and  $V_2$  have disjoint ranges, the greedy step over  $[V_1, V_2]$  runs the  $V_1$  step on inputs  $V_1$  reaches and the  $V_2$  step on inputs  $V_2$  reaches



|     |                                                                                                                                                                             |                                                                                                                                                                                                                                          |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{[V_1, V_2]^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2]$                                                                                                         | $\frac{[V_1, V_2]^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2] = (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2])$ |
| (b) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2])$      | $V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{\exists} = \text{ran}(V_1) \ \frac{V_1^\circ}{\exists} P(l)$                                                                                             |
| (c) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} P(l) \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2])$ | $V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} = \text{ran}(V_2) \ \frac{V_2^\circ}{\exists} P(r)$                                                                                             |



|     |                                                                                                                                                                                  |                                                    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| (a) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} P(l) \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} P(r) \text{est}(Q_1 + Q_2) [U_1, U_2])$ | $P(l) \text{est}(Q_1 + Q_2) = \text{est}(Q_1) \ l$ |
| (b) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1) \ l [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} P(r) \text{est}(Q_1 + Q_2) [U_1, U_2])$        | $P(r) \text{est}(Q_1 + Q_2) = \text{est}(Q_2) \ r$ |



|     |                                                                                                                                                                    |                |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| (a) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1) \ l [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_2) \ r [U_1, U_2])$ | $l [R, S] = R$ |
| (b) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1) \ U_1) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_2) \ r [U_1, U_2])$          | $r [R, S] = S$ |
| (c) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1) \ U_1) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_2) \ U_2)$                   |                |

## §10.2. The detab-entab problem

(10.2a)

$\text{detab} \triangleq ([\text{nil}, \text{expand}]) : \text{String} \rightarrow \text{String}$  over  $\text{snoc}$ -lists,  $\alpha \triangleq [\text{nil}, \text{snoc}]$ ,  $H = \text{detab}^\circ$ ;  $\text{expand } (xs, a) = (a = \text{TB} \rightarrow \text{fill } xs, xs \# [a])$ ,  $\text{fill } xs = xs \# \text{blanks } (n - (\text{col } xs) \bmod n)$ .

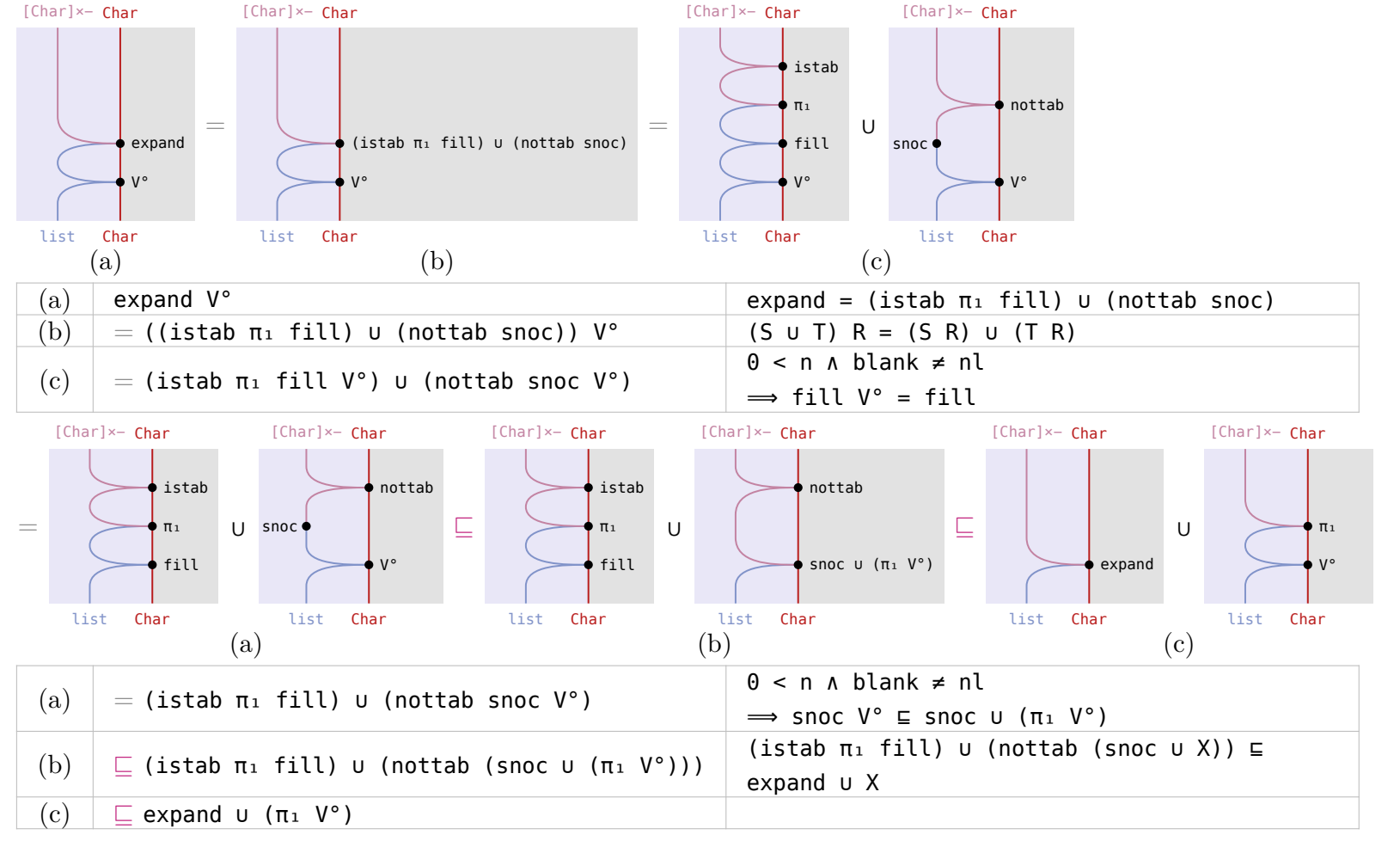
$\text{col} \triangleq ([\text{zero}, \text{count}])$ ,  $\text{count } (c, a) = (a = \text{NL} \rightarrow 0, c + 1)$ ; TB the tab, BL the blank, NL the newline, tab stops every  $n$  columns.

$R \triangleq \text{length} \leq \text{length}^\circ$ ,  $U$  the preorder with  $a \ U \ b \iff a = \text{TB} \vee a = b$ ,  $V \triangleq \text{prefix}^\circ n (\text{fill } \text{fill}^\circ)$ ,  $Q \triangleq 1 + (V \times U)$ .

$\text{unfill } xs$  the shortest prefix of  $xs$  with  $\text{fill}$  ( $\text{unfill } xs = \text{fill } xs$ );  $\text{tbc}$  the trailing blank count,  $\text{triple} \triangleq (\text{unfill } \text{entab}, (\text{tbc}, \text{col}))$ .

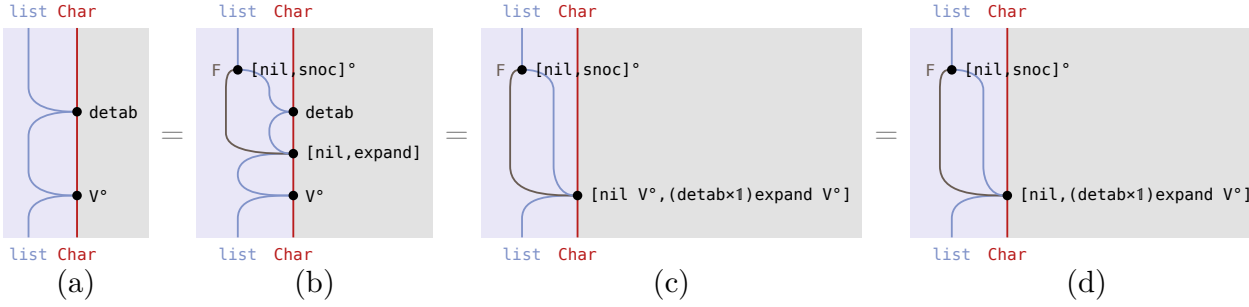
$$0 < n \wedge \text{blank} \neq \text{nl} \implies \text{expand } V^\circ \sqsubseteq \text{expand } u \ (\pi_1 V^\circ) \quad (10.2b)$$

shortening the output of one **expand** step to a  $V$ -smaller string either keeps the whole step (**expand**) or drops its character and shortens its input string ( $\pi_1 V^\circ$ )

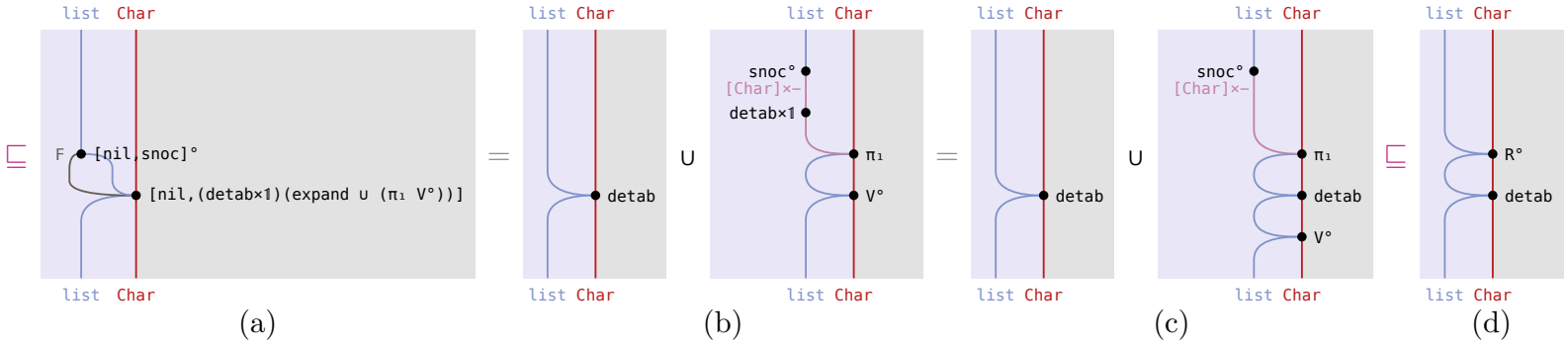


$$0 < n \wedge \text{blank} \neq n\mathbf{l} \Rightarrow \text{detab } V^\circ \sqsubseteq R^\circ \text{ detab} \quad (10.2c)$$

any  $V$ -shortening of **detab**'s output is **detab**'s output on an input no longer than the given one



|     |                                                                                                   |                                                                                              |
|-----|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| (a) | $\text{detab } V^\circ$                                                                           | $\text{detab} = [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand}]$ |
| (b) | $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand}] V^\circ$           | $[R, S] Z = [R Z, S Z]$                                                                      |
| (c) | $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil } V^\circ, \text{expand } V^\circ]$ | $\text{nil } V^\circ = \text{nil}$                                                           |
| (d) | $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } V^\circ]$          | (10.2b)                                                                                      |

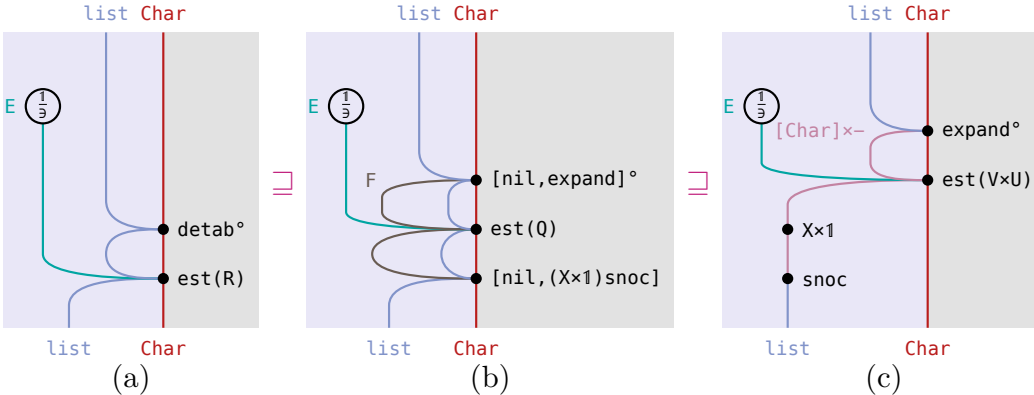


|     |                                                                                                              |                                                                                                                                                                  |
|-----|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\sqsubseteq [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } u (\pi_1 V^\circ)]$ | $[\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } u X] = \text{detab } u (\text{snoc}^\circ (\text{detab} \times 1) X)$               |
| (b) | $= \text{detab } u (\text{snoc}^\circ (\text{detab} \times 1) \pi_1 V^\circ)$                                | $\text{snoc}^\circ (\text{detab} \times 1) \pi_1 Y = \text{snoc}^\circ \pi_1 \text{ detab } Y$                                                                   |
| (c) | $= \text{detab } u (\text{snoc}^\circ \pi_1 \text{ detab } V^\circ)$                                         | $0 < n \wedge \text{blank} \neq n\mathbf{l}$<br>$\Rightarrow \text{detab } u (\text{snoc}^\circ \pi_1 \text{ detab } V^\circ) \sqsubseteq R^\circ \text{ detab}$ |
| (d) | $\sqsubseteq R^\circ \text{ detab}$                                                                          |                                                                                                                                                                  |

$$0 < n \wedge \text{blank} \neq n\mathbf{l} \Rightarrow (\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{3} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{detab}^\circ}{3} \text{ est}(R) \quad (10.2d)$$

the shortest input **detab** expands to the given output is one pass along that output, holding each blank back and caching the held blanks in for a tab wherever the column reaches a tab stop

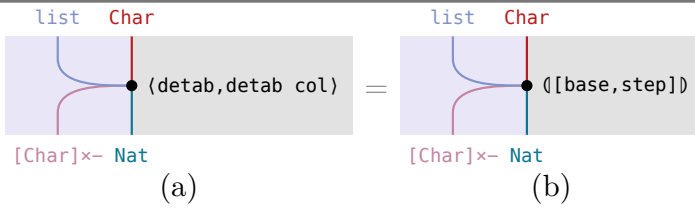
the specification —  $\triangleq \text{detab}$ ;  $\text{detab entab} = 1$  and nothing shorter does



|     |                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{\text{detab}^\circ}{3} \text{ est}(R)$                                                           | at $Q \triangleq 1 + (V \times U)$ : one character of input is decided at each step. $F(T, R) \alpha \sqsubseteq \alpha R$ . $\text{detab prefix} \sqsubseteq R^\circ \text{ detab}$ is FALSE, — at $n=8$ , $\text{detab}$<br>$[a, b, c, d, e, \text{TB}] = [a, b, c, d, e, \text{BL}, \text{BL}, \text{BL}]$ , whose prefix $[a, b, c, d, e, \text{BL}, \text{BL}]$ is longer than any input giving it, and $\text{detab } V^\circ \sqsubseteq R^\circ \text{ detab}$ holds. $\text{expand } V^\circ \sqsubseteq \text{expand } u (\pi_1 V^\circ)$ — shortening the output either leaves the last step alone or discards it |
| (b) | $\sqsubseteq \frac{[\text{nil}, \text{expand}]^\circ}{3} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}]$ | Proposition 10.1: <b>nil</b> and <b>expand</b> have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| (c) | $\sqsubseteq \frac{\text{expand}^\circ}{3} \text{ est}(V \times U) (X \times 1) \text{ snoc}$           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

$$\text{blank} \neq \text{nl} \wedge \text{tb} \neq \text{nl} \Rightarrow \langle \text{detab}, \text{detab col} \rangle = \langle [\text{base}, \text{step}] \rangle \quad (10.2e)$$

one pass over the input carries the output so far together with its column



|     |                                                  |                                                                                                                                                                                                  |
|-----|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle \text{detab}, \text{detab col} \rangle$ | both components are folds over the same input; <b>step</b> on $(\text{detab } x, \text{col}(\text{detab } x))$ is $(\text{detab}, \text{col detab})$ one character further — not a tabulated row |
| (b) | $= \langle [\text{base}, \text{step}] \rangle$   |                                                                                                                                                                                                  |

Exercise 10.1

- $\text{blank} \neq \text{nl} \wedge \text{tb} \neq \text{nl} \Rightarrow \langle [\text{base}, \text{step}] \rangle (\text{convert}(l), r) \Leftrightarrow r = \text{loop}(\text{step}, ([], 0), l)$

the fold of  $[\text{base}, \text{step}]$  over the input converted to a snoc-list is the left loop of **step** from **base**

- $\pi_1(\text{loop}(\text{step}, ([], c_0), x)) = \text{loop}'((f, g), (c_0, x))$

Exercise 10.1: when **step** only appends  $f(c, a)$  to the output and moves the state by  $g$ , the output of the loop is  $\text{loop}'(f, g)$ , which never carries the output

- $a \neq \text{tb} \Rightarrow \text{expand}(y, b) = \text{snoc}(x, a) \Leftrightarrow \text{fill}(y) = \text{snoc}(x, a) \wedge b = \text{tb} \vee (y, b) = (x, a)$

**expand** produces  $x\#[a]$  from a tab after a string that fills to it, or from  $x$  and  $a$

- $0 < n \wedge \text{blank} \neq \text{nl} \Rightarrow (\exists y. \text{fill}(y) = \text{snoc}(x, a)) \Leftrightarrow a = \text{blank} \wedge \text{col}(\text{snoc } x \text{ } a) \% n = 0$

a string is a **fill** exactly when it ends in a blank on a tab stop

- $\text{entab}(x) = \text{entab}(\text{unfill } x) ++ \text{blanks}(\text{tbc } x)$

(10.1): the output of **entab** is that of **unfill**  $x$  followed by the trailing blanks held back

- $\text{tbc}[] = 0$

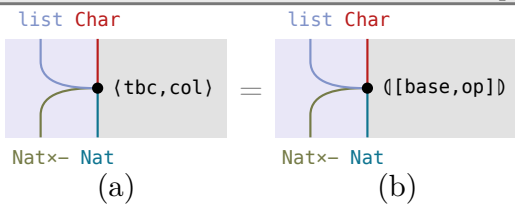
the empty string has no trailing blanks

- $\text{tbc}(\text{snoc}(x, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc } x \text{ } a) \% n \neq 0 \text{ then } \text{tbc } x + 1 \text{ else } 0$

a blank off a tab stop adds one held blank; anything else releases them all

$$\text{blank} \neq \text{nl} \Rightarrow \langle \text{tbc}, \text{col} \rangle = \langle [\text{base}, \text{op}] \rangle \quad (10.2g)$$

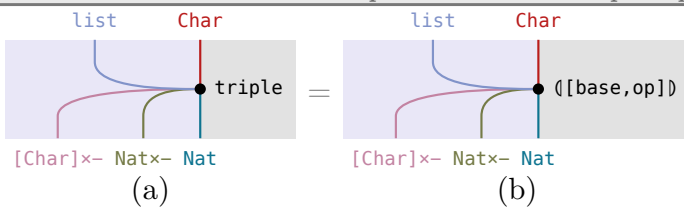
one pass counts the held blanks and the column together



|     |                                              |                                                                                                                                               |
|-----|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle \text{tbc}, \text{col} \rangle$     | <b>op</b> on $(\text{tbc } x, \text{col } x)$ is $(\text{tbc}, \text{col})$ one character further, from the equations for <b>tbc</b> — (10.1) |
| (b) | $= \langle [\text{base}, \text{op}] \rangle$ |                                                                                                                                               |

$$\text{blank} \neq \text{nl} \Rightarrow \text{triple} = \langle [\text{base}, \text{op}] \rangle \quad (10.2h)$$

one pass carries the output up to the held blanks, their count and the column



|     |                                              |                                                                                                                      |
|-----|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| (a) | <b>triple</b>                                | <b>op</b> on <b>triple</b> $x$ is <b>triple</b> one character further: (10.1) where the blanks are released — (10.1) |
| (b) | $= \langle [\text{base}, \text{op}] \rangle$ |                                                                                                                      |

| $\text{entab} = \text{triple assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$ <span style="float: right;">(10.2i)</span>                                                                                 |                                                                      |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------|
| entab is triple with the held blanks appended to its output                                                                                                                                                  |                                                                      |                 |
| <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>list Char</p> <p>(a)</p> </div> <div style="text-align: center;"> <p>list Char</p> <p>(b)</p> </div> </div> |                                                                      |                 |
| (a)                                                                                                                                                                                                          | entab                                                                | (10.1) — (10.1) |
| (b)                                                                                                                                                                                                          | $= \text{triple assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$ |                 |

### §10.3. The minimum tardiness problem

(10.3a)

$\text{FX} = 1 + (\text{X} \times \text{Job})$ ,  $\alpha \triangleq [\text{nil}, \text{snoc}]$  on schedules,  $\beta \triangleq [\text{nil}, \text{snag}]$  on bags, snag putting a job into a bag;  $\text{bagify} \triangleq \langle \beta \rangle : [\text{Job}] \rightarrow \text{Bag Job}$ ,  $\text{H} = \text{bagify}^\circ$ .

$\text{ct}, \text{dt}, \text{wt} : \text{Job} \rightarrow \text{Real}$  the completion, due and weighting quantities of a job;  $\text{penalty}(\text{xs}, \text{j}) = (\text{sum}(\text{list}(\text{ct})(\text{xs})) + \text{ct}(\text{j}) - \text{dt}(\text{j})) \times \text{wt}(\text{j})$ .

$\text{cost} \triangleq \frac{\text{prefix}}{\sup} \text{P}(\alpha^\circ [\text{zero}, \text{penalty}]) \text{ est}(\geq)$ ,  $\text{cost} [] = 0$ ,  $\text{cost} (\text{xs} + [\text{j}]) = \text{bmax} (\text{cost xs}, \text{penalty} (\text{xs}, \text{j}))$ ,  $\text{R} \triangleq \text{cost} \leq \text{cost}^\circ$ .

$\text{perm} \triangleq \text{bagify bagify}^\circ = \langle [\text{nil}, \text{add}] \rangle$ ,  $\text{add} (\text{xs}, \text{j}) = \text{ys} + [\text{j}] + \text{zs}$  for some  $\text{xs} = \text{ys} + \text{zs}$ .

$\text{k} \triangleq [\text{zero}, \text{assocr} (1 \times ((\text{bagify}^\circ \times 1) \text{ penalty})) \text{ bmax}]$ ,  $\text{f} \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{ penalty}]$ ,  $\text{Q} \triangleq \text{f} \leq \text{f}^\circ$ ,  $\text{Q}' \triangleq (\text{bagify}^\circ \times 1) \text{ penalty} \leq \text{penalty}^\circ (\text{bagify} \times 1)$ .

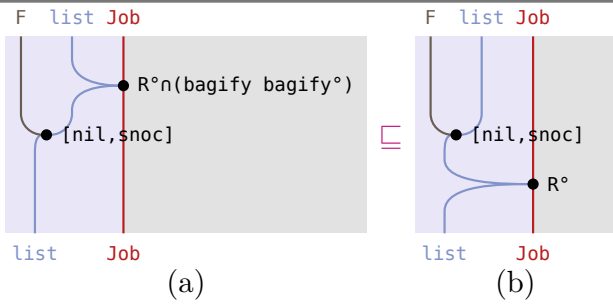
$\text{g} \triangleq [\text{zero}, \text{penalty}]$  (10.5),  $\text{m} \triangleq [\text{zero}, \pi_1 \text{ cost}]$  (B&dM's h, renamed as in (10.3e)) (10.6).

| $[\text{nil}, \text{snoc}] \text{ cost} = \text{F}(\langle \text{cost}, \text{bagify} \rangle) \text{ k}$ <span style="float: right;">(10.3b)</span>                                                              |                                                                    |                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the cost of a schedule is got from the cost and the bag of the schedule before its last job, and that job                                                                                                         |                                                                    |                                                                                                                                                                                                                           |
| <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>F [Job]x- Job</p> <p>(a)</p> </div> <div style="text-align: center;"> <p>F list Job</p> <p>(b)</p> </div> </div> |                                                                    |                                                                                                                                                                                                                           |
| (a)                                                                                                                                                                                                               | $[\text{nil}, \text{snoc}] \text{ cost}$                           | $\text{cost} (\text{xs} + [\text{j}]) = \text{bmax} (\text{cost xs}, \text{penalty} (\text{xs}, \text{j}))$ ,<br>and $\text{penalty}(\text{xs}, \text{j})$ reads only the bag of $\text{xs}$ — $\triangleq \text{bagify}$ |
| (b)                                                                                                                                                                                                               | $= \text{F}(\langle \text{cost}, \text{bagify} \rangle) \text{ k}$ |                                                                                                                                                                                                                           |

| $\text{F}(\geq \times 1) \text{ k} \sqsubseteq \text{k} \geq$ <span style="float: right;">Exercise 10.5</span>                                                                                                           |                                     |                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------|
| a larger cost before the last job gives a larger cost after it                                                                                                                                                           |                                     |                                                 |
| <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>F Intx- bag Job</p> <p>(a)</p> </div> <div style="text-align: center;"> <p>F Intx- bag Job</p> <p>(b)</p> </div> </div> |                                     |                                                 |
| (a)                                                                                                                                                                                                                      | $\text{F}(\geq \times 1) \text{ k}$ | $\text{bmax}$ is monotone in its first argument |
| (b)                                                                                                                                                                                                                      | $\sqsubseteq \text{k} \geq$         |                                                 |

$$F(R^\circ \sqcap (\text{bagify bagify}^\circ)) [\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}] R^\circ \quad (10.2)$$

improving the schedule before the last job, among schedules of the same bag, improves the whole schedule

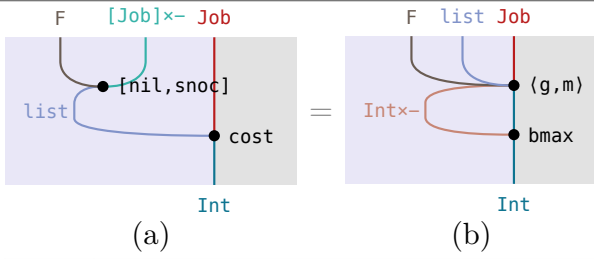


- |     |                                                                            |
|-----|----------------------------------------------------------------------------|
| (a) | $F(R^\circ \sqcap (\text{bagify bagify}^\circ)) [\text{nil}, \text{snoc}]$ |
| (b) | $\sqsubseteq [\text{nil}, \text{snoc}] R^\circ$                            |

Proposition 9.3 at  $S \triangleq \text{bagify}$ ,  $\leq \triangleq \geq$  — Proposition 9.3, with (10.3b) and Exercise 10.5

$$[\text{nil}, \text{snoc}] \text{ cost} = \langle g, m \rangle \text{ bmax} \quad (10.4)$$

the cost of a schedule is the larger of the penalty of its last job and the cost of the schedule before it

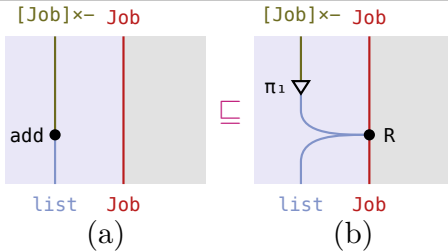


- |     |                                          |
|-----|------------------------------------------|
| (a) | $[\text{nil}, \text{snoc}] \text{ cost}$ |
| (b) | $= \langle g, m \rangle \text{ bmax}$    |

$\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$   
and  $\text{bmax}$  commutes —  $\triangleq \text{bagify}$

$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow \text{add} \sqsubseteq \pi_1 R \quad (10.7)$$

putting one more job anywhere into a schedule never lowers its cost

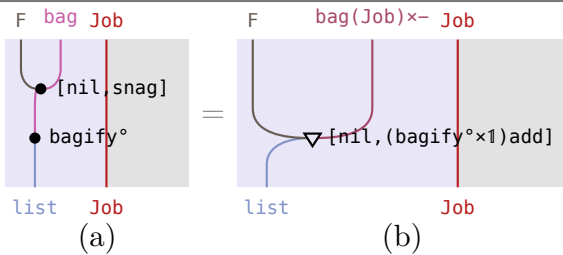


- |     |                       |
|-----|-----------------------|
| (a) | $\text{add}$          |
| (b) | $\sqsubseteq \pi_1 R$ |

induction on where  $j$  goes: every job after it starts later, so no penalty falls

$$[\text{nil}, \text{snag}] \text{ bagify}^\circ = [\text{nil}, (\text{bagify}^\circ \times 1) \text{ add}] \quad (10.8)$$

the orderings of a bag with one more job are the orderings of the bag with that job put in anywhere

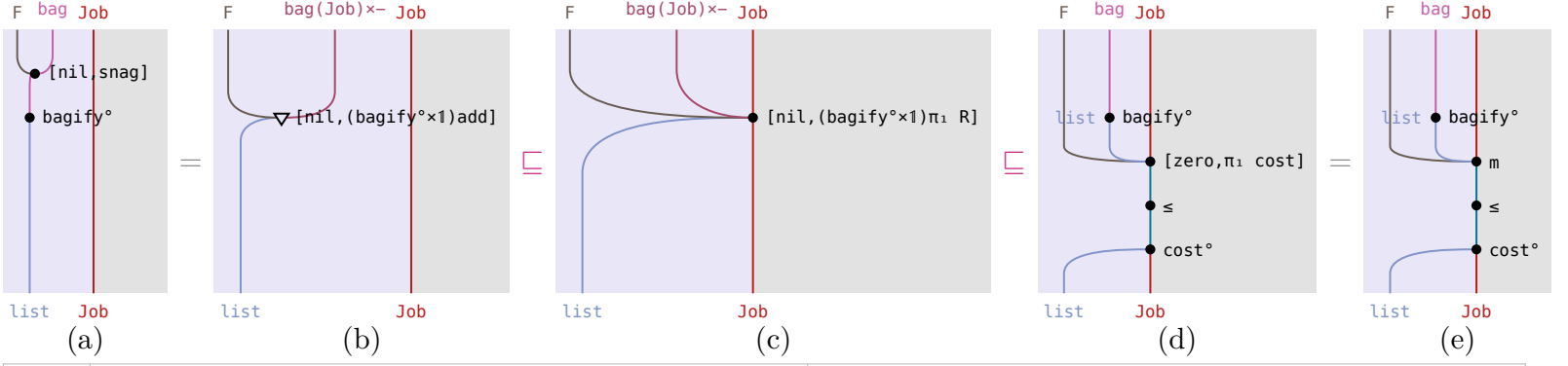


- |     |                                                              |
|-----|--------------------------------------------------------------|
| (a) | $[\text{nil}, \text{snag}] \text{ bagify}^\circ$             |
| (b) | $= [\text{nil}, (\text{bagify}^\circ \times 1) \text{ add}]$ |

take out the last copy of  $j$  and put it back —  $\triangleq \text{bagify}$

$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow [nil, snag] \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) m \leq \text{cost}^\circ \quad (10.3c)$$

an ordering of a bag with one more job costs at least the cost of the ordering of the bag before its last job

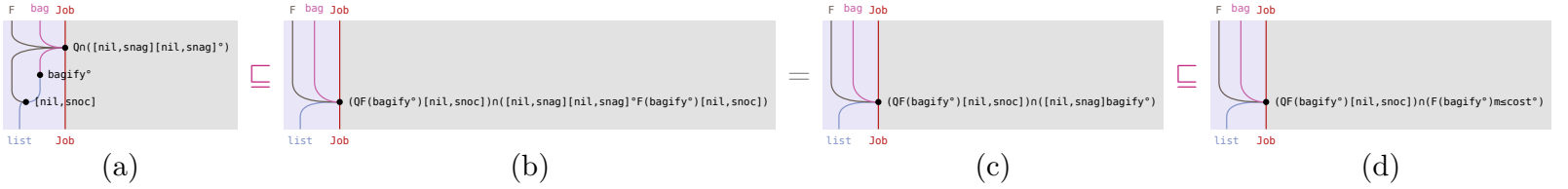


|     |                                                                                        |                                                                                |
|-----|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| (a) | $[nil, snag] \text{ bagify}^\circ$                                                     | (10.8)                                                                         |
| (b) | $= [nil, (\text{bagify}^\circ \times 1) \text{ add}]$                                  | (10.7)                                                                         |
| (c) | $\sqsubseteq [nil, (\text{bagify}^\circ \times 1) \pi_1 R]$                            | $[nil, \pi_1 R] \sqsubseteq [zero, \pi_1 \text{ cost}] \leq \text{cost}^\circ$ |
| (d) | $\sqsubseteq F(\text{bagify}^\circ) [zero, \pi_1 \text{ cost}] \leq \text{cost}^\circ$ |                                                                                |
| (e) | $= F(\text{bagify}^\circ) m \leq \text{cost}^\circ$                                    |                                                                                |

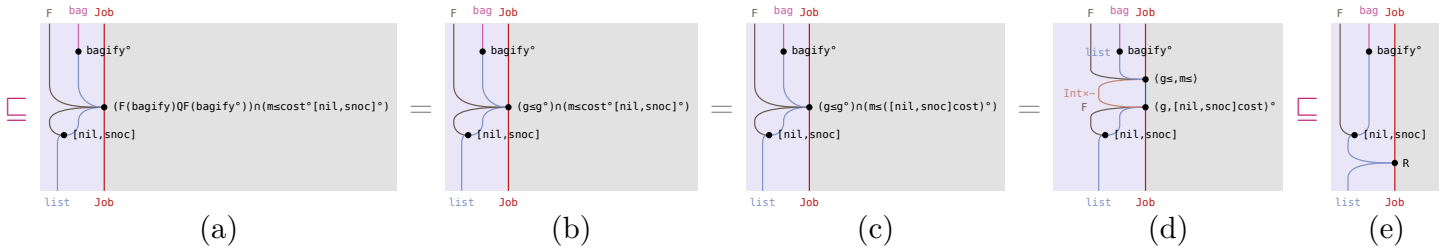
$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow (Q \cap ([nil, snag] [nil, snag]^\circ)) F(\text{bagify}^\circ) [nil, snoc] \sqsubseteq \quad (10.3)$$

$$F(\text{bagify}^\circ) [nil, snoc] R$$

a schedule of a bag ending in a job of least penalty is no worse than any schedule of the same bag



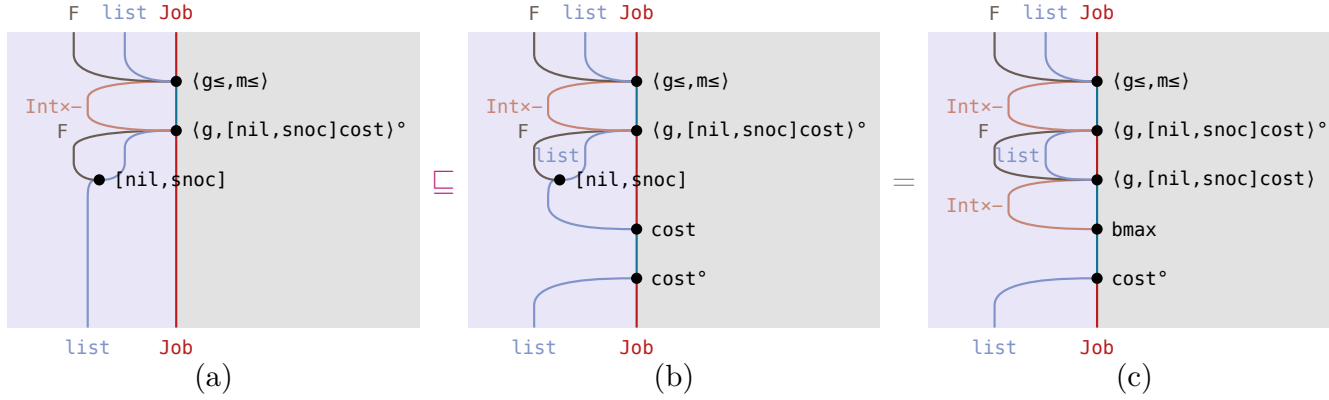
|     |                                                                                                                              |                                                                                                                                                                                                        |
|-----|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $(Q \cap ([nil, snag] [nil, snag]^\circ)) F(\text{bagify}^\circ) [nil, snoc]$                                                | $(S \cap T) R \sqsubseteq (S R) \cap (T R)$                                                                                                                                                            |
| (b) | $\sqsubseteq (Q F(\text{bagify}^\circ) [nil, snoc]) \cap ([nil, snag] [nil, snag]^\circ F(\text{bagify}^\circ) [nil, snoc])$ | $[nil, snag]^\circ F(\text{bagify}^\circ) [nil, snoc] = \text{bagify}^\circ$                                                                                                                           |
| (c) | $= (Q F(\text{bagify}^\circ) [nil, snoc]) \cap ([nil, snag] \text{ bagify}^\circ)$                                           | (10.3c)                                                                                                                                                                                                |
| (d) | $\sqsubseteq (Q F(\text{bagify}^\circ) [nil, snoc]) \cap (F(\text{bagify}^\circ) m \leq \text{cost}^\circ)$                  | $(Q F(\text{bagify}^\circ) [nil, snoc]) \cap (F(\text{bagify}^\circ) M) \sqsubseteq F(\text{bagify}^\circ) ((F(\text{bagify}^\circ) Q F(\text{bagify}^\circ)) \cap (M [nil, snoc]^\circ)) [nil, snoc]$ |



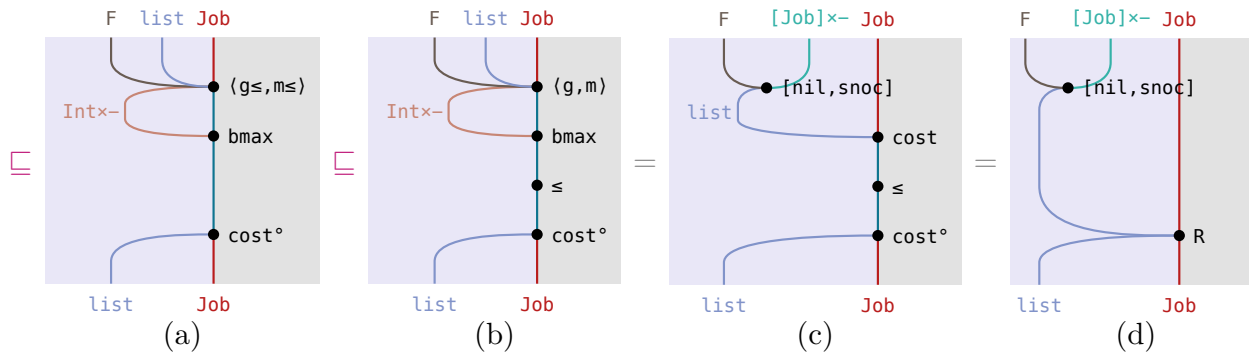
|     |                                                                                                                                                        |                                                                                  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| (a) | $\sqsubseteq F(\text{bagify}^\circ) ((F(\text{bagify}^\circ) Q F(\text{bagify}^\circ)) \cap (m \leq \text{cost}^\circ [nil, snoc]^\circ)) [nil, snoc]$ | $F(\text{bagify}^\circ) Q F(\text{bagify}^\circ) = g \leq g^\circ$               |
| (b) | $= F(\text{bagify}^\circ) ((g \leq g^\circ) \cap (m \leq \text{cost}^\circ [nil, snoc]^\circ)) [nil, snoc]$                                            | $(R S)^\circ = S^\circ R^\circ$                                                  |
| (c) | $= F(\text{bagify}^\circ) ((g \leq g^\circ) \cap (m \leq ([nil, snoc] \text{ cost}^\circ)^\circ)) [nil, snoc]$                                         | $\langle X, Y \rangle \langle R, S \rangle^\circ = (X R^\circ) \cap (Y S^\circ)$ |
| (d) | $= F(\text{bagify}^\circ) \langle g \leq, m \leq \rangle \langle g, [nil, snoc] \text{ cost}^\circ \rangle^\circ [nil, snoc]$                          | (10.3d)                                                                          |
| (e) | $\sqsubseteq F(\text{bagify}^\circ) [nil, snoc] R$                                                                                                     |                                                                                  |

$$\langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ [\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}] R \quad (10.3d)$$

a job whose penalty is at most the last penalty, put after a schedule costing at most the cost before the last job,  
gives a schedule costing at most the whole



|     |                                                                                                                                                                                      |                                                                                                                                           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ [\text{nil}, \text{snoc}]$                                                           | $1 \sqsubseteq \text{cost cost}^\circ$                                                                                                    |
| (b) | $\sqsubseteq \langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ [\text{nil}, \text{snoc}] \text{ cost cost}^\circ$                       | $[\text{nil}, \text{snoc}] \text{ cost} = \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle \text{ bmax}$                         |
| (c) | $= \langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle \text{ bmax cost}^\circ$ | $\langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle \sqsubseteq 1$ |



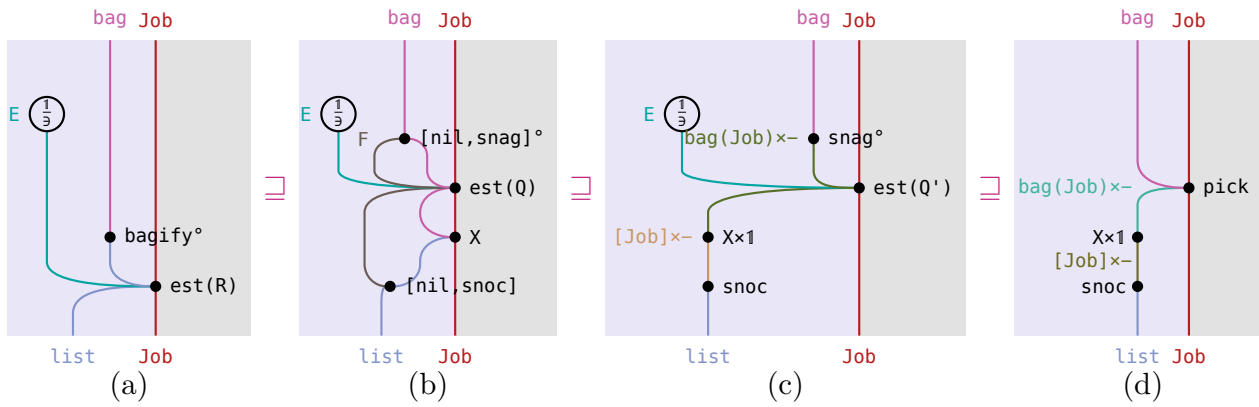
|     |                                                                        |                                                                                                  |
|-----|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| (a) | $\sqsubseteq \langle g \leq, m \leq \rangle \text{ bmax cost}^\circ$   | $\langle g \leq, m \leq \rangle \text{ bmax} \sqsubseteq \langle g, m \rangle \text{ bmax} \leq$ |
| (b) | $\sqsubseteq \langle g, m \rangle \text{ bmax} \leq \text{cost}^\circ$ | (10.4)                                                                                           |
| (c) | $= [\text{nil}, \text{snoc}] \text{ cost} \leq \text{cost}^\circ$      | $R = \text{cost} \leq \text{cost}^\circ$                                                         |
| (d) | $= [\text{nil}, \text{snoc}] R$                                        |                                                                                                  |

$$\text{schedule} = (\text{null} \rightarrow \text{nil}, \text{pick} (\text{schedule} \times 1) \text{ snoc}), (\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (\mu X : \quad (10.3e)$$

$$\frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}] \sqsubseteq \frac{\text{bagify}^\circ}{\exists} \text{ est}(R)$$

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag,  
putting it last, and scheduling what is left the same way

the specification —  $\triangleq \text{bagify}$



|     |                                                                                                             |                                                                                                                                                                    |
|-----|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{\text{bagify}^\circ}{\exists} \text{ est}(R)$                                                        | No greedy <b>reduce</b> exists — one would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole |
| (b) | $\sqsubseteq \frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}]$ | Proposition 10.1: <b>nil</b> and <b>snag</b> have disjoint ranges                                                                                                  |
| (c) | $\sqsubseteq \frac{\text{snag}^\circ}{\exists} \text{ est}(Q') (X \times 1) \text{ snoc}$                   | $\text{pick} \sqsubseteq \frac{\text{snag}^\circ}{\exists} \text{ est}(Q')$ , a partial function, quadratic in the number of jobs                                  |
| (d) | $\sqsubseteq \text{pick} (X \times 1) \text{ snoc}$                                                         |                                                                                                                                                                    |

## §10.4. The TeX problem

(10.4a)

$\text{interval} \triangleq \text{val round} : \text{Decimal} \rightarrow [0, 2^{16})$ ,  $\text{val} \triangleq ([\text{zero}, \text{shift}])$ ,  $\text{shift } (d, r) = (d+r)/10$ ,  $\text{round } r$  rounds  $2^{16}r$  to the nearest integer:  $\text{round } r = n \iff 2n-1 < 2^{17}r < 2n+1$ .

$\text{interval } n = ((2n-1)/2^{17}, (2n+1)/2^{17})$ ,  $r \text{ inrange } (a, b) \iff a < r < b$ ,  $\text{round}^\circ = \text{interval inrange}$ ,  $R \triangleq \text{length} \leq \text{length}^\circ$ .

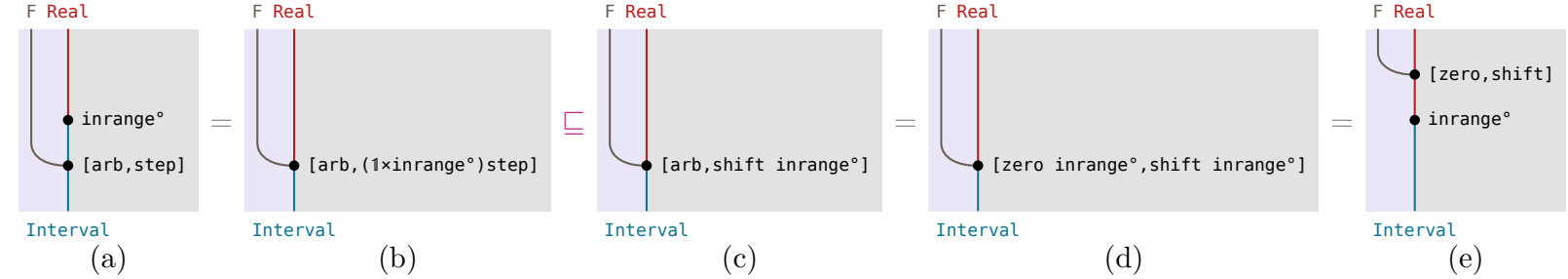
Interval the pairs  $(a, b)$  with  $\text{Legal}(a, b) \iff 0 < b < 1$  and  $a < b$  (10.9);  $\text{step\_legal}$  is what types  $\text{step } [\text{arb}, \text{step}] : 1 + (\text{Digit} \times \text{Interval}) \rightarrow \text{Interval}$ ,  $\text{step } (d, (a, b)) = ((d+a)/10, (d+b)/10)$ .

$\text{FX} = 1 + (\text{Digit} \times X)$ ,  $\alpha \triangleq [\text{nil}, \text{cons}]$ ,  $H \triangleq ([\text{arb}, \text{step}])^\circ$ ,  $! : \text{Digit} \times \text{Interval} \rightarrow 1$ ,  $Q \triangleq (l^\circ !^\circ r) \cup 1$   $l, r$  are (5.3a.1)'s injections into  $\text{FX} = 1 + (\text{Digit} \times X)$ , so  $l : 1 \rightarrow \text{FX}$  and  $r : \text{Digit} \times X \rightarrow \text{FX}$ ,  $w \triangleq 2^{17}$ .

$([\text{arb}, \text{step}])^\circ \sqsubseteq \text{val inrange}^\circ$

(10.4b)

every interval got by folding a decimal's digits with  $[\text{arb}, \text{step}]$  has the decimal's value strictly inside it

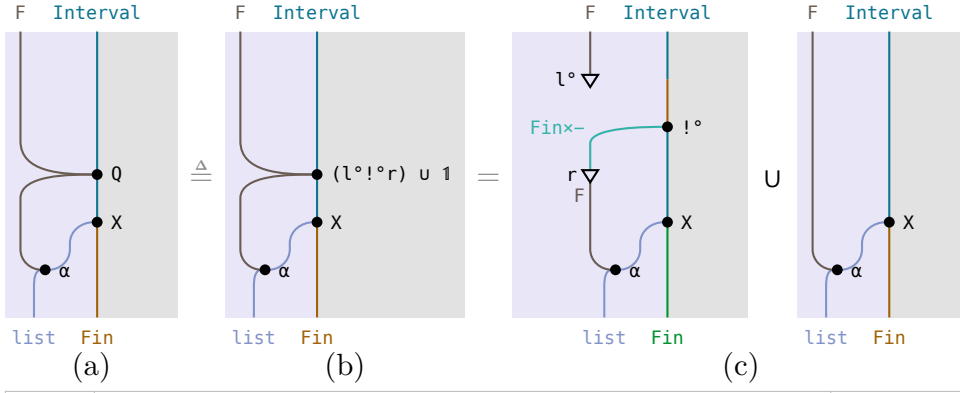


|     |                                                               |                                                                                      |
|-----|---------------------------------------------------------------|--------------------------------------------------------------------------------------|
| (a) | $F(\text{inrange}^\circ) [\text{arb}, \text{step}]$           | $F(S) [T, U] = [T, (1 \times S) U]$                                                  |
| (b) | $= [\text{arb}, (1 \times \text{inrange}^\circ) \text{step}]$ | $(1 \times \text{inrange}^\circ) \text{step} \sqsubseteq \text{shift inrange}^\circ$ |
| (c) | $\sqsubseteq [\text{arb}, \text{shift inrange}^\circ]$        | $\text{zero inrange}^\circ = \text{arb}$                                             |
| (d) | $= [\text{zero inrange}^\circ, \text{shift inrange}^\circ]$   | $[R, S] Z = [R Z, S Z]$                                                              |
| (e) | $= [\text{zero}, \text{shift}] \text{ inrange}^\circ$         |                                                                                      |

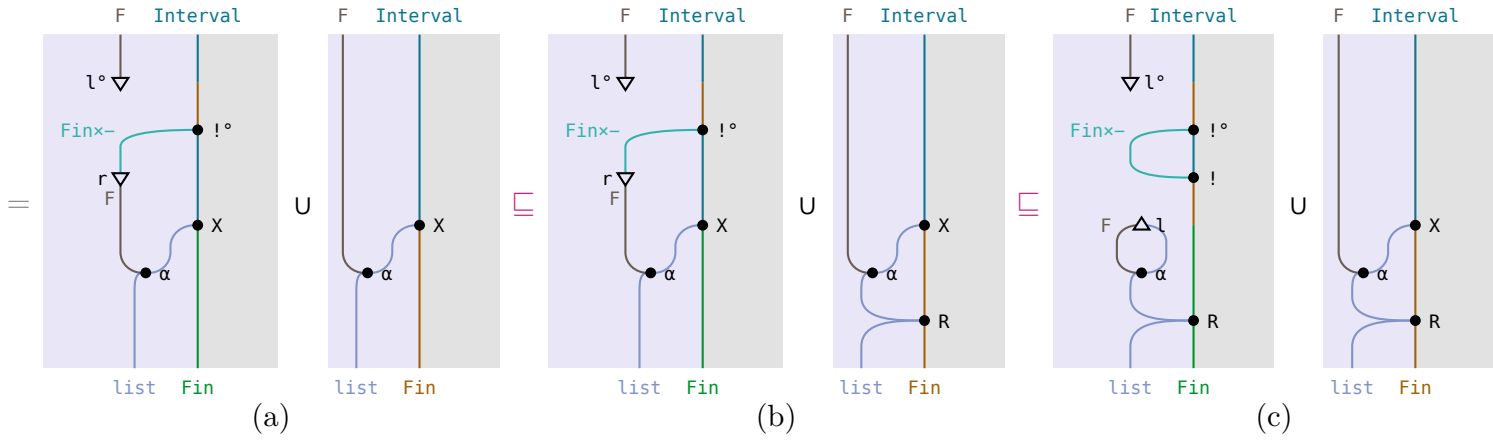
$$Q F(X) \alpha \sqsubseteq F(X) \alpha R$$

(10.4c)

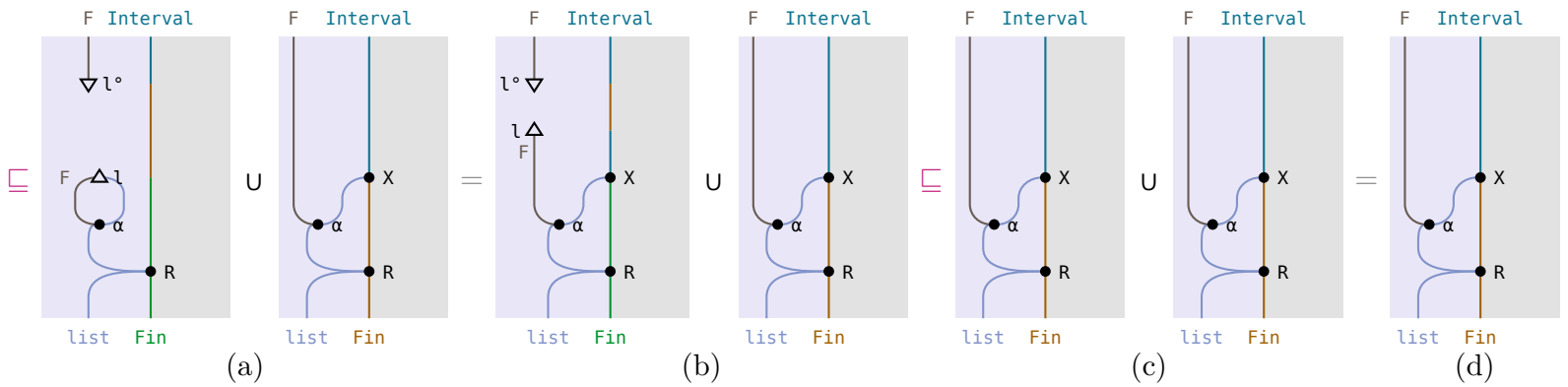
choosing by  $Q$  among the one-step decompositions before building with  $F(X)$  and  $\alpha$  yields only decimals that are  $R$ -related to one built without choosing



|     |                                             |                                |
|-----|---------------------------------------------|--------------------------------|
| (a) | $Q F(X) \alpha$                             | $Q \triangleq (l^o !^o r) u 1$ |
| (b) | $\triangleq ((l^o !^o r) u 1) F(X) \alpha$  | $(S u T) R = (S R) u (T R)$    |
| (c) | $= (l^o !^o r F(X) \alpha) u (F(X) \alpha)$ |                                |



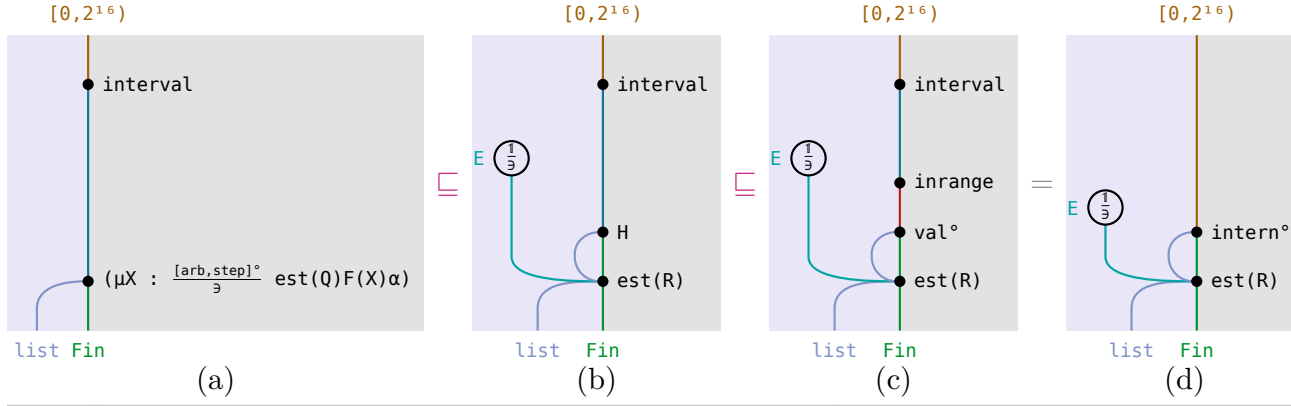
|     |                                                         |                                          |
|-----|---------------------------------------------------------|------------------------------------------|
| (a) | $= (l^o !^o r F(X) \alpha) u (F(X) \alpha)$             | $1 \sqsubseteq R$                        |
| (b) | $\sqsubseteq (l^o !^o r F(X) \alpha) u (F(X) \alpha R)$ | $r F(X) \alpha \sqsubseteq ! l \alpha R$ |
| (c) | $\sqsubseteq (l^o !^o ! l \alpha R) u (F(X) \alpha R)$  | $!^o ! \sqsubseteq 1$                    |



|     |                                                        |                           |
|-----|--------------------------------------------------------|---------------------------|
| (a) | $\sqsubseteq (l^o !^o ! l \alpha R) u (F(X) \alpha R)$ | $l F(X) = l$              |
| (b) | $= (l^o !^o ! l F(X) \alpha R) u (F(X) \alpha R)$      | $l^o !^o l \sqsubseteq 1$ |
| (c) | $\sqsubseteq (F(X) \alpha R) u (F(X) \alpha R)$        | $R u R = R$               |
| (d) | $= F(X) \alpha R$                                      |                           |

$$\text{extern}(n)=f(2n-1,2n+1), \quad \text{interval } (\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q) F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{est}(R) \quad (10.4d)$$

a shortest decimal whose internal representation is the given multiple of  $2^{-16}$  is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do



|     |                                                                                                  |                                                                                                                                     |
|-----|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{interval } (\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q) F(X) \alpha)$ | $\frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q) F(\frac{H}{3} \text{est}(R)) \alpha \sqsubseteq \frac{H}{3} \text{est}(R)$ |
| (b) | $\sqsubseteq \text{interval } \frac{H}{3} \text{est}(R)$                                         | $\frac{H}{3} \text{est}(R) \sqsubseteq \frac{\text{inrange } \text{val}^\circ}{3} \text{est}(R)$                                    |
| (c) | $\sqsubseteq \text{interval } \frac{\text{inrange } \text{val}^\circ}{3} \text{est}(R)$          | $\frac{\text{intern}^\circ}{3} = \text{interval } \frac{\text{inrange } \text{val}^\circ}{3}$                                       |
| (d) | $= \frac{\text{intern}^\circ}{3} \text{est}(R)$                                                  |                                                                                                                                     |

- $f(a,b) = x \iff a < 0 \wedge x = [] \vee \neg a < 0 \wedge (\exists d. \exists y. d = [10b] \wedge f(10a-d, 10b-d) = y \wedge x = [d] \# y)$   
f gives (a,b) the empty decimal when  $a < 0$ , and otherwise the digit  $d = [10b]$  followed by what it gives  $(10a-d, 10b-d)$

(10.4e)

- $(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q) F(X) \alpha) = f$

the least solution of the greedy recursion of (10.4d) is f

- $d = [10b] \wedge e = [10b] \implies d = e$

Exercise 10.12: two digits d and e with  $0 < 10b-d < 1$  and  $0 < 10b-e < 1$  are equal

- $(0 < q-p \wedge q < w) \wedge p < 0 \implies f(p,q) = []$

the program's first clause:  $f(p,q) = []$  when  $p < 0$

- $(0 < q-p \wedge q < w) \wedge \neg p < 0 \wedge d = 10q/w \implies f(p,q) = [d] \# f(10p-wd, 10q-wd)$

the program's second clause: with  $d = (10q) \text{ div } w$ ,  $f(p,q)$  is d followed by  $f(10p-w \cdot d, 10q-w \cdot d)$

- $(0 < q-p \wedge q < w) \wedge \text{Reach}(p,q) \implies f(p/w, q/w) = f(p,q)$

on the pairs the program reaches from interval n, the integer f at (p,q) is a value of the rational f at  $(p/w, q/w)$

- $\text{interval } f = \text{extern}$

f after interval is the program extern, whose value at n is  $f(2n-1, 2n+1)$

## §10. Appendix

### §10.1. $P(S) \text{ est}(R) = (\exists S) n(\epsilon \setminus (SR^\circ))$

(10.1a)

$(p, q)$  tabulates  $W \triangleq (\exists S) n(\epsilon \setminus (SR^\circ))$ , and  $y \triangleq \frac{(p \exists S) n(qR)}{\exists}$ , a map.

(10.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} W \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = W, 1 \sqsubseteq yy^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(10.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f^\circ, \cdot \exists \neg \exists} \boxed{\begin{array}{c} q \\ (p \exists S) n(qR) \end{array}} \xleftarrow{\Delta \neg n, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

$$\boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \exists, \circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple}$$

(10.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg n, \cdot T \neg / T, \circ, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array}$$

$$\boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \exists, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple}$$

$$\boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) n(qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq qq^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} W \\ \epsilon \setminus (SR^\circ) \end{array}} \quad \text{W's right half}$$

### §10.2. $P(\text{est}(R)) \text{ est}(R) = P(\text{dom}(\text{est}(R))) \text{ union est}(R)$

(10.2a)

$$\begin{array}{c} \boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)} \\ \xrightarrow{P \text{ a relator}} \boxed{P(\text{dom}(\text{est}(R))) P(\text{est}(R)) \text{ est}(R)} \xrightarrow{(7.11)} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \end{array}$$

(10.2b)

$$\begin{array}{c} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ P(\text{est}(R)) \text{ est}(R) \end{array}} \xleftarrow{(10.1b) \text{ at } S := \text{est}(R), \Delta \neg n, T \cdot \neg T \setminus} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \boxed{\begin{array}{c} \epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \end{array}$$

$$\begin{array}{c} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \xleftarrow{P(\text{dom}(\text{est}(R))) \sqsubseteq 1} \boxed{\begin{array}{c} \text{union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \xleftarrow{T(U \cap V) \sqsubseteq T U \cap T V, \cdot \exists \neg \exists, (7.3)} \boxed{\begin{array}{c} (\exists \exists) n(\epsilon \setminus (\epsilon \setminus R^\circ)) \\ \exists \text{est}(R) \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} \exists (\exists n(\epsilon \setminus (\epsilon \setminus R^\circ))) \\ \exists (\exists n(\epsilon \setminus R^\circ)) \end{array}} \quad \text{count of } T \cdot \neg T \setminus \\ \boxed{\begin{array}{c} \epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\begin{array}{c} \text{dom}(\text{est}(R)) \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \\ \xleftarrow{\text{dom}(\text{est}(R)) \sqsubseteq \text{est}(R) \text{ est}(R)^\circ} \boxed{\begin{array}{c} \text{est}(R) \text{ est}(R)^\circ \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{est}(R)^\circ \sqsubseteq \epsilon, \epsilon \epsilon \text{ union} \sqsubseteq \epsilon} \boxed{\begin{array}{c} \text{est}(R) \epsilon \text{est}(R) \\ \text{est}(R) R^\circ \end{array}} \quad \text{UP of est} \end{array}$$

### §10.3. Shortest paths on a cylinder, on lists

(10.3a)

|    | definition                                                                                                         | type                                               | note                                                                                                                                   |
|----|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1  | $F(A, X) = A + A \times X$                                                                                         | $\mathcal{A} \rightarrow \mathcal{A}$              | The base functor: one square of the new column, alone or in front of the path so far.                                                  |
| 2  | $L = \text{list}^+$                                                                                                | $\mathcal{A} \rightarrow \mathcal{A}$              | A path is a non-empty list of squares, one per column crossed.                                                                         |
| 3  | $\alpha$ the initial algebra                                                                                       | $F(\text{dCL}(L, E)) \rightarrow \text{dCL}(L, E)$ | $\alpha(5) = [5]$ and $\alpha(1, [5]) = [1, 5]$ : a path is started by one square, or extended by one.                                 |
| 4  | $N$<br>the n-tuple relator                                                                                         | $\mathcal{A} \rightarrow \mathcal{A}$              | One component per row of a column, so the fold carries n answers at once.                                                              |
| 5  | $R \triangleq \text{sum} \leq \text{sum}^\circ$                                                                    | $[\text{Nat}] \rightarrow [\text{Nat}]$            | The cost of a path, which the cheapest minimises.                                                                                      |
| 6  | setify                                                                                                             | $A[n] \rightarrow PA$                              | $\text{setify}(1, 2, 3, 4) = \{1, 2, 3, 4\}$ — which row a component came from is forgotten.                                           |
| 7  | moves                                                                                                              | $A[n] \rightarrow P(A[n])$                         | $\text{moves}(x) = \{\text{up}(x), x, \text{down}(x)\}$ — rotated up, unrotated, rotated down.                                         |
| 8  | trans                                                                                                              | $P(A[n]) \rightarrow (P A)[n]$                     | $\text{trans}\{(a, b, c), (x, y, z)\} = (\{a, x\}, \{b, y\}, \{c, z\})$ — component k of the result is the set of the k-th components. |
| 9  | zip                                                                                                                | $F(a[n]) \rightarrow (F a)[n]$                     | $\text{zip}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = ((1, \{[5]\}), (2, \{[6]\}), (3, \{[7]\}), (4, \{[8]\}))$ .          |
| 10 | $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$                                                               | $F(PA) \rightarrow PFA$                            | $\text{cp}(1, \{[5], [6], [8]\}) = \{(1, [5]), (1, [6]), (1, [8])\}$ .                                                                 |
| 11 | $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union}))$<br>$\text{zip } N(\text{cp}(F(A, -)) E(\alpha))$ | $F(NA, N(P(TA))) \rightarrow N(P(TA))$             | $\text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}))$ is worked out in (10.3.1b).                                           |
| 12 | $\text{paths} \triangleq (\text{gen}) \text{ setify union}$                                                        | $T(NA) \rightarrow P(TA)$                          | $\text{paths}[(1, 2, 3, 4), (5, 6, 7, 8)]$ is the union of (10.3.1b)'s four sets: 12 paths, 3 from each entry row.                     |
| 13 | the specification<br>$\text{paths est}(R)$                                                                         | $[\text{Nat}[n]] \rightarrow [\text{Nat}]$         | A cheapest path from the entry side to the exit side.                                                                                  |

§10.3.1.  $N(E(L A))$ 

1 5  
 2 6  
 3 7       $[(1, 2, 3, 4), (5, 6, 7, 8)] : L \ N \ \text{Nat}$   
 4 8

(10.3.1a)

$(\text{gen})[(5, 6, 7, 8)] = (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$

(10.3.1b)

$\text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) =$   
 $(\{[1, 5], [1, 6], [1, 8]\},$   
 $\{[2, 5], [2, 6], [2, 7]\},$   
 $\{[3, 6], [3, 7], [3, 8]\},$   
 $\{[4, 5], [4, 7], [4, 8]\})$

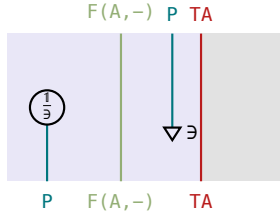
(10.3.1c)

|   |          |                                                        |
|---|----------|--------------------------------------------------------|
| 1 | <b>N</b> | the 4-tuple, one component per row of the entry column |
| 2 | <b>E</b> | the set of paths that can start in that row            |
| 3 | <b>L</b> | a path is the non-empty list of the squares it crosses |

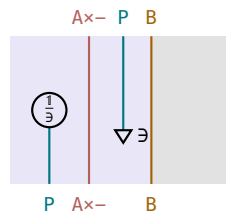
row 1's set has no  $[1,7]$  — from row 1 only rows 4, 1 and 2 are reachable — and it does have  $[1,8]$ , because the cylinder glues the bottom row to the top. That is **moves trans N(union)** of (10.3a.11): component **k** collects the paths of rows **k-1**, **k**, **k+1**.

### §10.3.2. The cross product $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$

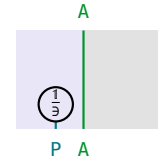
(10.3.2a)



cp on all of  $F(A, E(L A))$



the  $A \times -$  summand:  $\exists$  picks one path,  $1\% \exists$  collects the results



the **A** summand, no **E** to distribute:  $1\% \exists$  alone,  $a \mapsto \{a\}$

(10.3.2b)

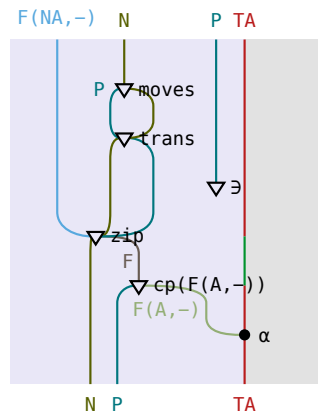
| type                                                                | note                                                                                                   |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 1 $F(A, -) = A + A \times -$                                        | one wire carries the whole functor, both summands with it; the $\exists$ under it is $F(1, \exists)$   |
| 2 $\exists : E(L A) \rightarrow L A$                                | the $\exists$ inside $F(1, \exists)$ : a set of paths, one of them — $L A$ is one path, $A$ one square |
| 3 $F(1, \exists) = 1 + 1 \times \exists$                            | $: A + A \times E(L A) \rightarrow A + A \times L A$ , the relator acting on each summand              |
| 4 $\exists : E(A + A \times L A) \rightarrow A + A \times L A$      | the $\exists$ under the bar: a set of cells, one of them — a different $\exists$                       |
| 5 $\text{cp} : A + A \times E(L A) \rightarrow E(A + A \times L A)$ | $\frac{R}{\exists}$ turns $R : X \rightarrow Y$ into $X \rightarrow E Y$                               |
| 6 $\text{cp} = \frac{1}{\exists} E(F(1, \exists))$                  | what the picture draws: the unit makes the outer $E$ , leaving one $\exists$ — (0.10a.13)              |

(10.3.2c)

$w = (1, \{[5], [6], [8]\}) : A \times E(L A)$  the right summand  
 $1 \times \exists : A \times E(L A) \rightarrow A \times L A$   $1$  keeps the square 1,  $\exists$  picks one path  
 $w \mapsto (1, [5]), (1, [6]), (1, [8]) : A \times L A$  one output per path in  $w$   
 $\text{cp}(w) = \{(1, [5]), (1, [6]), (1, [8])\} : E(A + A \times L A)$  the three of them, collected

### §10.3.3. $\text{gen} = F(1, \text{moves trans N(union)}) \text{ zip N(cp P}(\alpha))$

(10.3.3a)

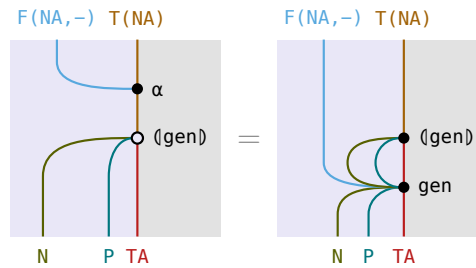


(10.3.3b)

$u = ((1,2,3,4),(\{[5]\},\{[6]\},\{[7]\},\{[8]\}))$  :  $F(N A, N(E(L A)))$   
 $F(1, \text{moves trans } N(\text{union}))$  : 1 keeps the column, the path SETS move  
 $\text{moves}(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$   
 $= (\{[6]\},\{[7]\},\{[8]\},\{[5]\}),$   
 $(\{[5]\},\{[6]\},\{[7]\},\{[8]\}),$   
 $(\{[8]\},\{[5]\},\{[6]\},\{[7]\})$  :  $E(N(E(L A)))$  down, unmoved, up  
 $\text{trans}(\text{that})$   
 $= (\{[6]\},\{[5]\},\{[8]\},\{[7]\},\{[6]\},\{[5]\}),$   
 $(\{[8]\},\{[7]\},\{[6]\},\{[5]\},\{[8]\},\{[7]\})$  :  $N(E(E(L A)))$  row k gets rows k-1, k, k+1  
 $N(\text{union})(\text{that})$   
 $= (\{[5],[6],[8]\},\{[5],[6],[7]\},$   
 $\{[6],[7],[8]\},\{[5],[7],[8]\})$  :  $N(E(L A))$  every path into row k, none dropped  
 $\text{zip}(\text{that})$   
 $= ((1,\{[5],[6],[8]\}), (2,\{[5],[6],[7]\}),$   
 $(3,\{[6],[7],[8]\}), (4,\{[5],[7],[8]\}))$  :  $N(F(A, E(L A)))$  each row: its square, and the paths it may be put in front of  
 $N(\text{cp } P(\alpha))(\text{that})$   
 $= (\{[1,5],[1,6],[1,8]\},\{[2,5],[2,6],[2,7]\},$   
 $\{[3,6],[3,7],[3,8]\},\{[4,5],[4,7],[4,8]\})$  :  $N(E(L A))$  cp pairs the square with each path,  $\alpha$  prefixes it:  $\alpha(1,[5])=[1,5]$

#### §10.3.4. $\text{gen}$ is an F-algebra; $(\text{gen}): \alpha(\text{gen})=F(1, (\text{gen}))\text{gen}$

(10.3.4a)

 $\alpha$  puts the column back on the list, then the fold reads all of itthe fold reads the rest under  $F$ , then one  $\text{gen}$  puts the column in front

$$xs = [(1,2,3,4), (5,6,7,8)] = \alpha((1,2,3,4), [(5,6,7,8)]) : L(N A)$$

(10.3.4b)

$$(\text{gen})(xs) = \text{gen}((1,2,3,4), (\text{gen})[(5,6,7,8)])$$

both halves are in (10.3.1b): the tail folds to  $(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$ , one path per row and each of them one square long, and  $\text{gen}$  on it is (10.3.3b)'s walk.

#### §10.3.5. $Q : F(N A, N(L A)) \rightarrow N(L A)$ , an $F(N A, -)$ -algebra: $Q=F(1, \text{moves trans } N(\text{est}(R))) \text{ zip } N(\alpha)$

(10.3.5a)

$Q : F(N A, N(L A)) \rightarrow N(L A)$  : the fold's algebra: a new column, one path per row  
 $u = ((1,2,3,4),([5],[6],[7],[8]))$  :  $F(N A, N(L A))$   
 $F(1, \text{moves trans } N(\text{est}(R)))$  : 1 keeps the column, the paths move  
 $\text{moves}([5],[6],[7],[8])$   
 $= (\{[6],[7],[8],[5]\},$   
 $\{[5],[6],[7],[8]\},$   
 $\{[8],[5],[6],[7]\})$  :  $E(N(L A))$  down, unmoved, up  
 $\text{trans}(\text{that})$   
 $= (\{[6],[5],[8]\},\{[7],[6],[5]\},$   
 $\{[8],[7],[6]\},\{[5],[8],[7]\})$  :  $N(E(L A))$  row k gets rows k-1, k, k+1  
 $N(\text{est}(R))(\text{that})$   
 $= ([5],[5],[6],[5])$  :  $N(L A)$  the cheapest into each row – chosen BEFORE the new square  
 $\text{zip}(\text{that})$   
 $= ((1,[5]), (2,[5]), (3,[6]), (4,[5]))$  :  $N(F(A, L A))$  each row: its square, and the one predecessor that survived  
 $N(\alpha)(\text{that})$   
 $= ([1,5],[2,5],[3,6],[4,5])$  :  $N(L A)$   $\alpha(1,[5])=[1,5]$

$$F(1, \text{moves trans } N(\text{est}(R))) = [l, pm \ r] \wedge \text{zip} = [N(\text{inl}), \text{zip}' \ N(\text{inr})] \wedge \text{inl } \alpha = \text{wrap} \wedge \text{inr } \alpha \quad (10.3.5b)$$

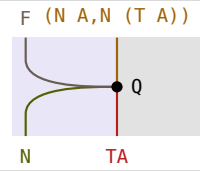
$$= \text{cons} \Rightarrow Q = [N(\text{wrap}), pm \ \text{zip}' \ N(\text{cons})]$$

at the last column  $Q$  starts one path per row, and at each earlier one it puts each square in front of the cheapest of the three kept paths it can step to — the algebra (10.3.5c)'s last step folds

**circuit** — the fork is  $F(NA, N(LA)) = NA + NA \times N(LA)$

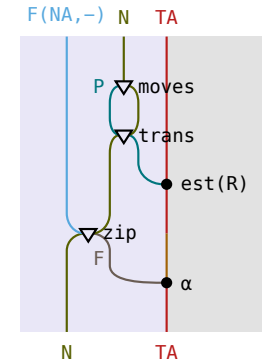
**Hinze–Marsden**

$$F(N \ A) \ (N \ (T \ A)) \rightarrow Q \rightarrow N(T \ A)$$



$$= FN(T \ A) \rightarrow F(1, \text{moves trans } N(\text{est}(R))) \rightarrow FN(T \ A) \rightarrow \text{zip} \rightarrow N(F \ A \ (T \ A)) \rightarrow N(\alpha) \rightarrow N(T \ A)$$

the fusion condition (10.3.5d) read as a definition,



$$= F(N \ A) \ (N \ (T \ A)) \rightarrow \left( \begin{array}{l} l \rightarrow N(p_0) \rightarrow N(\text{wrap}) \\ r \rightarrow q \rightarrow pm \rightarrow q \rightarrow \text{zip}' \rightarrow N(q_0) \rightarrow N(\text{cons}) \end{array} \right) \rightarrow N(T \ A)$$

$$\text{zip} = 1 + \text{zip}', \alpha = [\text{wrap}, \text{cons}]$$

